Does event-JEPA pretraining beat its own random initialisation?

A proposal on scaling HEPA/KITE

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1 Motivation

Scaling HEPA [1] or KITE [2] into a foundation model assumes the frozen pretrained encoder knows something the same architecture does not at initialisation; ROCKET [4] showed random features are hard to beat in time-series classification. I tested that assumption on the public code.

2 Pilot: pretrained vs. never-trained encoder

I ran `scripts/train.py` at its PROTOCOL defaults (d=256, 2.16M parameters, 50/30 epochs) with `--pre-epochs 50` and `--pre-epochs 0`; the latter leaves the encoder at initialisation (hash-checked) and runs the same finetuning. Pretrained FD001 reaches 0.901, against the ∼0.902 in `REPRODUCIBILITY.md`.

Table 1: Mean per-horizon AUROC (`h_auroc`), mean ± sd over 3 seeds; Δ is paired.

	FD001	FD003	MBA	PSM	ETTm1
Random encoder, never trained	$.924 \pm .002$	$.957 \pm .002$	$.799 \pm .051$	$.631 \pm .008$	$.812 \pm .010$
Pretrained encoder (published recipe)	$.901 \pm .007$	$.941 \pm .003$	$.820 \pm .010$	$.629 \pm .022$	$.809 \pm .011$
Δ (pretrained − random)	$-.0225 \pm .0063$	$-.0155 \pm .0056$	$+.0210 \pm .0430$	$-.0022 \pm .0139$	$-.0029 \pm .0172$
Seeds where pretrained is ahead	0/3	0/3	2/3	1/3	1/3

The deficit is in the encoder (Fig. 1a): a fresh predictor on the pretrained encoder keeps it; the pretrained predictor on a random encoder removes it. The targets collapse: their centred effective rank falls from about 13 to 1.3 over pretraining. A variance–covariance penalty on the targets raises it to 40–54 and closes most of the gap, but does not put pretraining ahead of random. Width does not help (Fig. 1b): pretraining trails random in 17 of 18 C-MAPSS seed × width runs. At a reduced budget, one encoder shared across all five datasets lost 0.017 mean pooled AUROC (0.823 vs. 0.840); the same target penalty recovered it (0.839, two seeds).

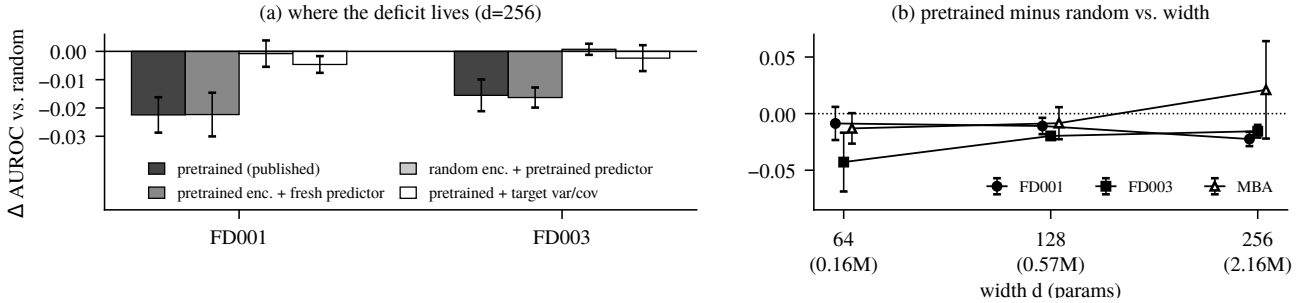

Figure 1: Per-horizon AUROC minus the never-trained encoder's (paired, mean ± sd, 3 seeds). (a) Encoder/predictor swap and target penalty. (b) Width sweep.

3 Hypothesis

Event-JEPA pretraining beats its own initialisation only when the target is kept from collapsing and the corpus spans many domains. At per-dataset scale it does not; as data and model grow together, the pretrained-minus-random gap turns positive only with a target-side regulariser. Refuted if (a) under the paper's configuration (detached target, SIGReg, 100/50 epochs) pretraining beats random by more than one seed-sd on most of the 14 benchmarks; (b) at 10–40M parameters on FactoryNet [3] plus the 14 benchmarks the gap stays ≤ 0 even with SIGReg on the targets; or (c) it turns positive with no target-side regulariser.

4 Plan

(1) *CPU, ∼6–8 core-hours:* detached-target configuration, pretraining-length sweep (0/5/15/50 epochs), FD002/FD004. (2) *One GPU, est. 20–60 A10G-hours:* all 14 benchmarks × 5 seeds × {published objective, random encoder, target var/cov, SIGReg, frozen target}, plus d=512 and 1024; outcome (a) ends the project here. (3) *Forgis compute, est. 190–750 GPU-hours:* FactoryNet plus the 14 benchmarks with KITE variate tokens; three objectives × three seeds × {2M, 10M, 40M}; the random control needs only finetunes. GPU estimates are extrapolated from 472 core-s/run on CPU; KITE's mean variate count is the least certain input. Whatever the outcome, a "same architecture, never trained" row costs one finetune and turns "beats baselines" into "pretraining added this much".

Limitations. CPU only (2 cores); five of fourteen benchmarks; three seeds on a fixed split. The public code is not the paper's run (VICReg-style regulariser rather than SIGReg, joint rather than detached target, 50/30 epochs), so none of this speaks to the published numbers. Both clear losses are on C-MAPSS (global z-scoring, no RevIN), which may favour random projections; MBA, PSM and ETTm1 get at most 50 pretraining steps.

Code and per-run results: github.com/Siddhant-Shankar/Siddhant_Shankar_HEPA_KITE_Proposal

[1] HEPA, arXiv:2605.11130; [2] KITE, arXiv:2610.12738; [3] FactoryNet, arXiv:2605.09081; [4] Dempster et al., ROCKET, DMKD 2020.